



A machine-learning forecast for Lake Michigan water temperature

Ryan McComb · June 2026 · buoy 45174, Evanston-Wilmette shoreline · not live yet, in final testing

Two kilometres off the Wilmette shore there is a NOAA buoy that reports the water temperature every ten minutes, May through November. Swimmers check it constantly, because this corner of Lake Michigan is moody: one hard north wind can tear the warm surface layer off the beach overnight and drop the water ten degrees while everyone sleeps.

Buoycast is my attempt to forecast that buoy. It is deliberately simple machinery: gradient-boosted decision trees (scikit-learn, no neural networks), trained on ten seasons of history, 2016 through 2026. Each training example pairs one moment of the lake, described by 46 numbers (recent water temperatures, wind, sun, air temperature, the season clock, and the weather forecast over the coming window) with what the water actually did afterward. That gives the model 599,000 examples to learn from.

It forecasts every hour out to seven days. Five copies of the model predict five percentiles, so the output is not a single number but a band; the whole thing is run across 34 different weather-model futures, so the band widens exactly when the weather models genuinely disagree; and every forecast is pinned to the buoy's live reading at hour one.

TRAINING EXAMPLES

INPUTS PER HOUR

SEASONS REPLAYED

VERIFIED TEST FORECASTS

REFRESH

599,000

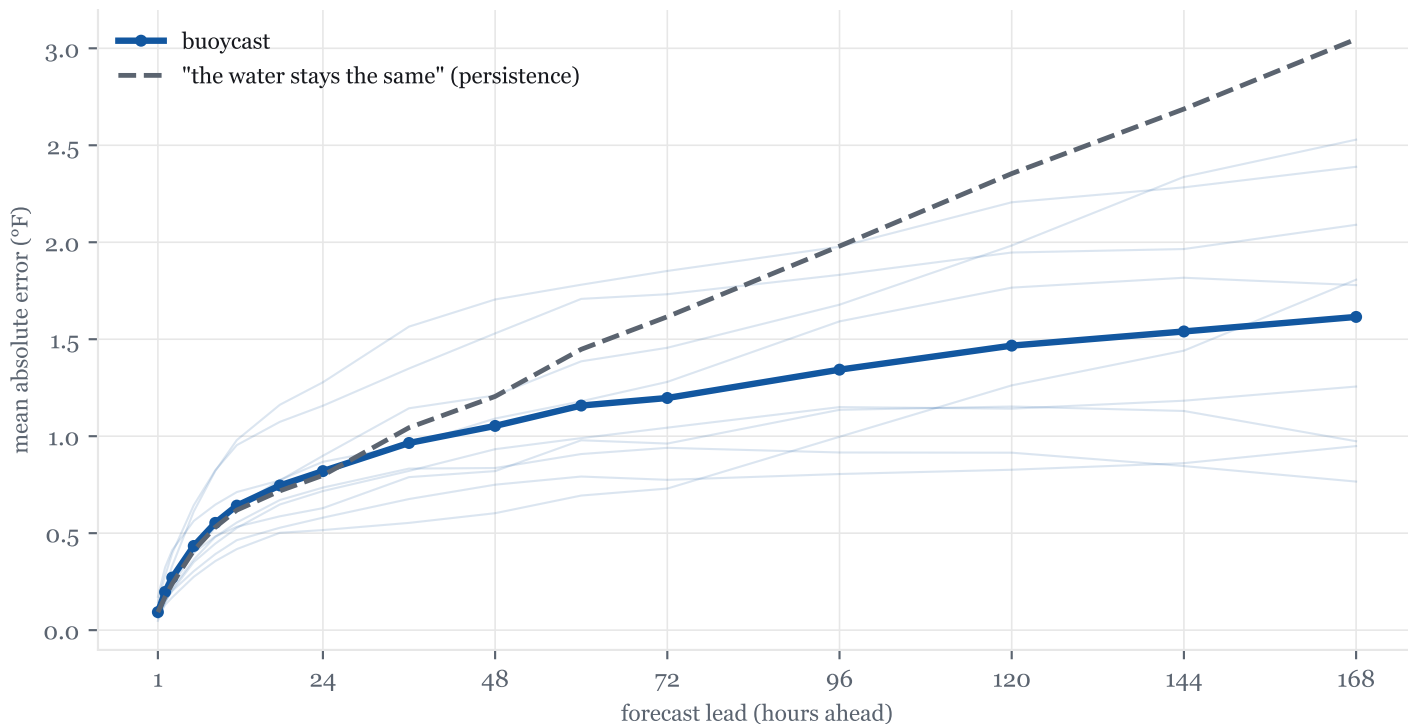
46

9

131,745

10 min

Figure 1 · How far off it is, by lead time: nine seasons replayed honestly

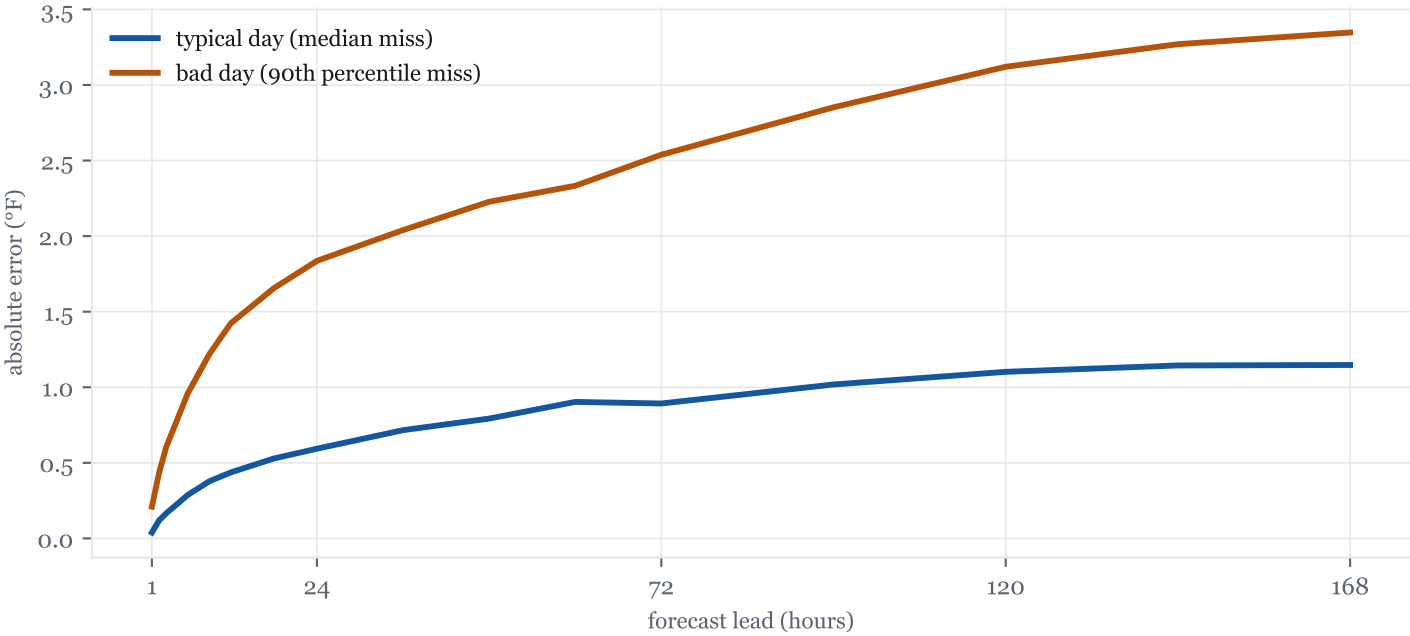


Every number comes from data the model had never seen: it was retrained nine times, once per season 2018-2026, and graded only on held-out weeks (131,745 forecast-versus-outcome pairs). Thin lines are individual seasons. At one day the lake barely moves, so beating persistence is nearly impossible; from two days out the model pulls away, and at a week it is roughly twice as accurate.

lead	model	"no change"	edge
+1 hour	0.09 °F	0.09 °F	tie
+1 day	0.82 °F	0.80 °F	tie
+3 days	1.20 °F	1.62 °F	+26%
+7 days	1.61 °F	3.05 °F	+47%

Half of all one-day forecasts miss by under half a degree. The published band is calibrated on those nine replayed seasons, so when it says 90 percent, it holds the real outcome at least 90 percent of the time. When the lake is about to do something violent, the band widens and says so.

Figure 2 · Typical day vs bad day. The gap is why the forecast carries a band



The hard days are real: autumn wind events (upwelling) can drop this shoreline 10 to 16 degrees in a day, and the worst one-day miss across nine seasons was 10.5 °F during exactly such a storm. The model's answer is honesty rather than bravado: those are the days its uncertainty band visibly opens up.

What happens next. The live version is in final testing: a dashboard that refreshes every ten minutes against the buoy, retrains itself every Sunday, and runs on one small cloud VM and open data end to end. It will be public soon.

Credit where it is due. None of these ideas are new; they are old, public ideas applied carefully. The validation approach, replaying whole hidden seasons and grading the model cold, runs parallel to the VoteHub 2026 midterm methodology, which is mainly the work of Zachary Donnini; most of what I know about checking a forecast honestly comes from that write-up and from questions Zachary has answered for me. Data: NOAA NDBC (the buoy), ERA5 reanalysis (Copernicus), and the GFS, ECMWF, ICON, and GEM weather models via Open-Meteo, all open and free.

Stack: Python, scikit-learn, pandas; one HistGradientBoostingRegressor per quantile; conformal calibration on pooled out-of-sample residuals; anchor-blended to the live observation.